

Soumya Ranjan Nayak

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OBJECTIVE

Computer Science undergraduate (Bachelor of Engineering, 2027) with hands-on experience building backend systems, real-time data pipelines, and applied machine learning models – ranging from anomaly detection to predictive maintenance. Looking to apply strong fundamentals in data structures, algorithms, and systems design to build reliable, scalable software.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, Go, SQL

Data Structures & Algorithms: Arrays, Graphs, DP, Binary Search, Two Pointers, DSU – 200+ problems solved

Backend & Systems: FastAPI, REST APIs, WebSocket, Docker, Git

AI/ML: PyTorch, Scikit-learn, XGBoost, LSTM, Ensemble Models, Real-time Anomaly Detection

Databases: PostgreSQL, MySQL, Redis | **Practices:** Agile, Unit Testing, Code Review, OOP

Soft Skills: Team Leadership (GDG chapter, 2,100+ members), Cross-functional Collaboration (6-member SIH team), Technical Communication (dashboards built for non-technical end users)

EDUCATION

Cambridge Institute of Technology North Campus

Bangalore, India

Bachelor of Engineering in Computer Science | CGPA: 8.3/10

Aug 2023 – Jun 2027

WORK EXPERIENCE

Machine Learning Intern

Sep – Nov 2025

Infosys Springboard

Bangalore, India

- Built an LSTM model to predict Remaining Useful Life (RUL) of aircraft engines from time-series sensor data (NASA C-MAPSS dataset), achieving $R^2 = 0.71$ and RMSE = 22.43 cycles.
- Designed a multi-level alert system (Emergency/Critical/Warning/Normal thresholds) and built an interactive Streamlit dashboard for fleet monitoring and analytics.

IoT Apprentice

Aug 2025 – May 2026

Samsung Innovation Campus

Bangalore, India

- Built a Raspberry Pi-based assistive technology system to support autism spectrum disorder (ASD) therapy, using GPIO sensor integration and automated logging for multi-user session tracking.
- Applied TCP/IP networking and debugged hardware-software integration issues between sensors and the Pi.

RESEARCH & PROJECTS

AEGIS – Cybercrime Anomaly Detection | *Python, XGBoost, FastAPI*

2025

- Grand Finalist, Smart India Hackathon 2025 (Ministry of Home Affairs) – ranked 1,360 of 1,42,715 teams (top 1%) in a 6-member team.
- Built an ensemble model (XGBoost + Isolation Forest + Gradient Boosting, ~90% accuracy) to flag suspicious transactions in simulated cybercrime complaint data, plus an LSTM model to predict money-flow patterns over time.
- Built a live, auto-refreshing dashboard to display risk scores and support faster investigation decisions.

Yantraraksha – Real-time ICS/SCADA Threat Detection | *Python, FastAPI, WebSocket*

2025

- Built the backend in FastAPI with 17 REST endpoints and a WebSocket layer to stream live security alerts to a dashboard.
- Trained an ML ensemble (Random Forest + Isolation Forest) reaching 98.92% precision across 5 categories of network attacks (mapped to MITRE ATT&CK for ICS).
- Integrated Nmap for periodic network scanning and automated Windows firewall rules to block detected threats.

ACHIEVEMENTS & LEADERSHIP

1st Place – Industry Innovation Hackathon 2026

Sathyabama Institute

Won outright among 110+ competing teams in a 24-hour sprint

Grand Finalist – Smart India Hackathon 2025

Ministry of Home Affairs

Rank 1,360 of 1,42,715 teams (top 1%)

Chapter Head – Google Developer Groups (GDG) On Campus

Cambridge Institute of Technology

Grew chapter to 2,100+ members; organized 8+ technical events

Competitive Programming: 200+ problems solved across LeetCode, CodeChef, and GeeksforGeeks | Finalist – ProtoVision Ignite 2024